lab1 : Data hiding in slack space and forensic analysis

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1/

```
┌──(black㉿kali)-[~]
└─$ sudo dd if=/dev/zero of=virtual_disk.img bs=1M count=10
[sudo] Mot de passe de black :
10+0 enregistrements lus
10+0 enregistrements écrits
10485760 octets (10 MB, 10 MiB) copiés, 0,00463305 s, 2,3 GB/s
```

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```
┌──(black㉿kali)-[~]
└─$ sudo losetup /dev/loop10 virtual_disk.img

┌──(black㉿kali)-[~]
└─$ sudo losetup -a
/dev/loop10: [2049]:1588932 (/home/black/virtual_disk.img)
```

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```
┌──(black㉿kali)-[~]
└─$ sudo mkfs.ext4 -F -b 4096 /dev/loop10
mke2fs 1.47.2 (1-Jan-2025)
Discarding device blocks: done
Creating filesystem with 2560 4k blocks and 2560 inodes

Allocating group tables: done
Writing inode tables: done
Creating journal (1024 blocks): done
Writing superblocks and filesystem accounting information: done
```

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```
┌──(black㉿kali)-[~]
└─$ sudo dumpe2fs -h /dev/loop10 | grep "Block size"
dumpe2fs 1.47.2 (1-Jan-2025)
Block size:               4096
```

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```
┌──(black㉿kali)-[~]
└─$ sudo mkdir -p /mnt/virtual_disk
```

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```
┌──(black㉿kali)-[~]
└─$ sudo mount /dev/loop10 /mnt/virtual_disk

┌──(black㉿kali)-[~]
└─$ df -h /mnt/virtual_disk
Sys. de fichiers Taille Utilisé Dispo Uti% Monté sur
/dev/loop10        5,4M    152K  4,6M   4% /mnt/virtual_disk
```

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```
┌──(black㉿kali)-[~]
└─$ sudo cp /usr/share/legion/images/hp-big.jpg /mnt/virtual_disk/plane.jpg
```

8/

```
┌──(black㉿kali)-[~]
└─$ ls -l /mnt/virtual_disk/plane.jpg
-rw-r--r-- 1 root root 5752  9 nov.  15:23 /mnt/virtual_disk/plane.jpg

┌──(black㉿kali)-[~]
└─$ SIZE=5752
SLACK=$(( (4096 - SIZE % 4096) % 4096 ))
echo "Slack space: $SLACK bytes"
Slack space: 2440 bytes
```

slack space = 2440 bytes

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```
┌──(black㉿kali)-[~]
└─$ sudo umount /mnt/virtual_disk
```

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```
┌──(black㉿kali)-[~]
└─$ sudo debugfs /dev/loop10
debugfs 1.47.2 (1-Jan-2025)
debugfs:  stat /plane.jpg
debugfs:
```

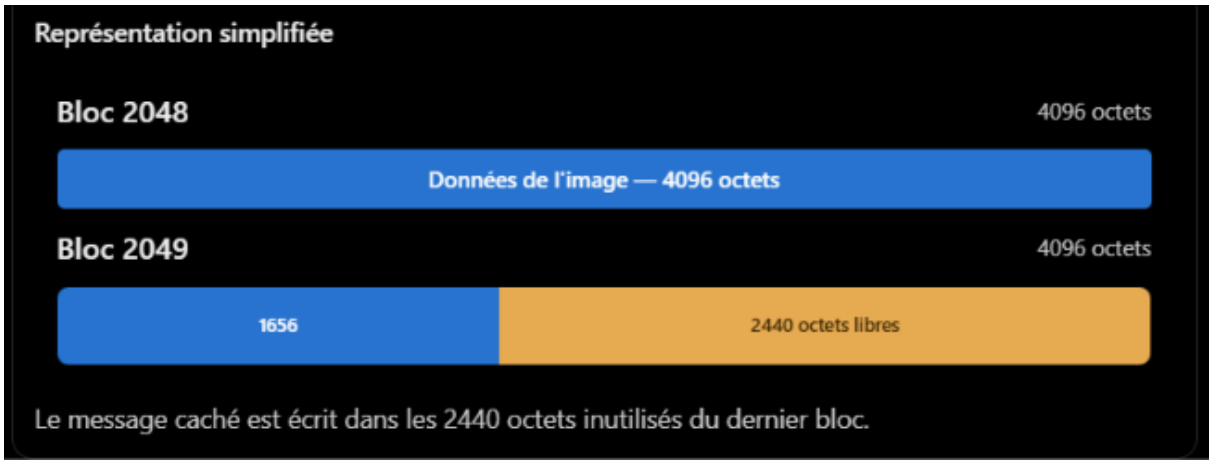
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```
Inode: 13   Type: regular    Mode:  0644   Flags: 0×80000
Generation: 533896339    Version: 0×00000000:00000001
User:     0   Group:     0   Project:     0   Size: 5752
File ACL: 0
Links: 1   Blockcount: 16
Fragment:  Address: 0    Number: 0    Size: 0
 ctime: 0×6af1d76a:63e94550 -- Mon Nov  9 15:23:38 2026
 atime: 0×6af1d76a:63e94550 -- Mon Nov  9 15:23:38 2026
 mtime: 0×6af1d76a:63e94550 -- Mon Nov  9 15:23:38 2026
crtime: 0×6af1d76a:63e94550 -- Mon Nov  9 15:23:38 2026
Size of extra inode fields: 32
Inode checksum: 0×46f490c8
EXTENTS:
(0-1):2048-2049
~
~
~
~
~
~
```

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The first block contains 4096 bytes, leaving 1656 bytes of the file in the second block.

Offset = 2049*4096 + 1656 = 8 394 360 bytes



13/

```
┌──(black㉿kali)-[~]
└─$ OFFSET=8394360
LAST_BLOCK=2049
INODE=13

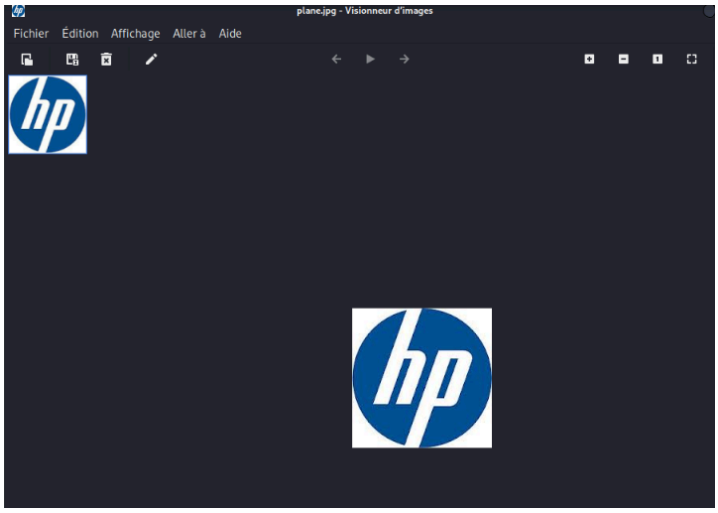
┌──(black㉿kali)-[~]
└─$ printf '%s' "this is a hidden message" | sudo dd of=/dev/loop10 bs=1 seek=$OFFSET conv=notrunc
24+0 enregistrements lus
24+0 enregistrements écrits
24 octets copiés, 0,00159064 s, 15,1 kB/s

┌──(black㉿kali)-[~]
└─$ sudo dd if=/dev/loop10 bs=1 skip=$OFFSET count=24 status=none
echo
this is a hidden message
```

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```
┌──(black㉿kali)-[~]
└─$ sudo mount /dev/loop10 /mnt/virtual_disk
```

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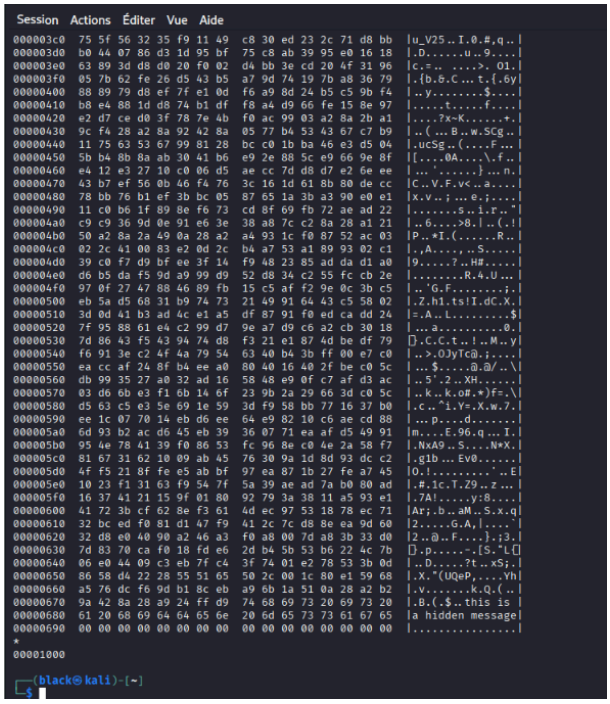
16/

```
┌──(black㉿kali)-[~]
└─$ strings /mnt/virtual_disk/plane.jpg | grep "hidden"
```

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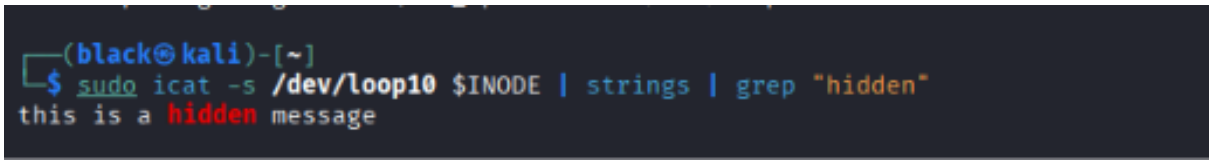
```
┌──(black㉿kali)-[~]
└─$ sudo hexdump -C -s $((OFFSET-32)) -n 96 /dev/loop10
00801658  50 2c 00 1c 80 e1 59 68  a5 76 dc f6 9d b1 8c eb  |P,....Yh.v......|
00801668  a9 6b 1a 51 0a 28 a2 b2  9a 42 8a 28 a9 24 ff d9  |.k.Q.( ...B.(.$..|
00801678  74 68 69 73 20 69 73 20  61 20 68 69 64 64 65 6e  |this is a hidden|
00801688  20 6d 65 73 73 61 67 65  00 00 00 00 00 00 00 00  | message........|
00801698  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  |................|
*
008016b8
```

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19/





Le message « this is a hidden message » a été récupéré avec succès. Cela montre que les données cachées dans le Slack Space, invisibles lors d'une lecture normale du fichier, peuvent être retrouvées à l'aide d'outils d'analyse forensic.